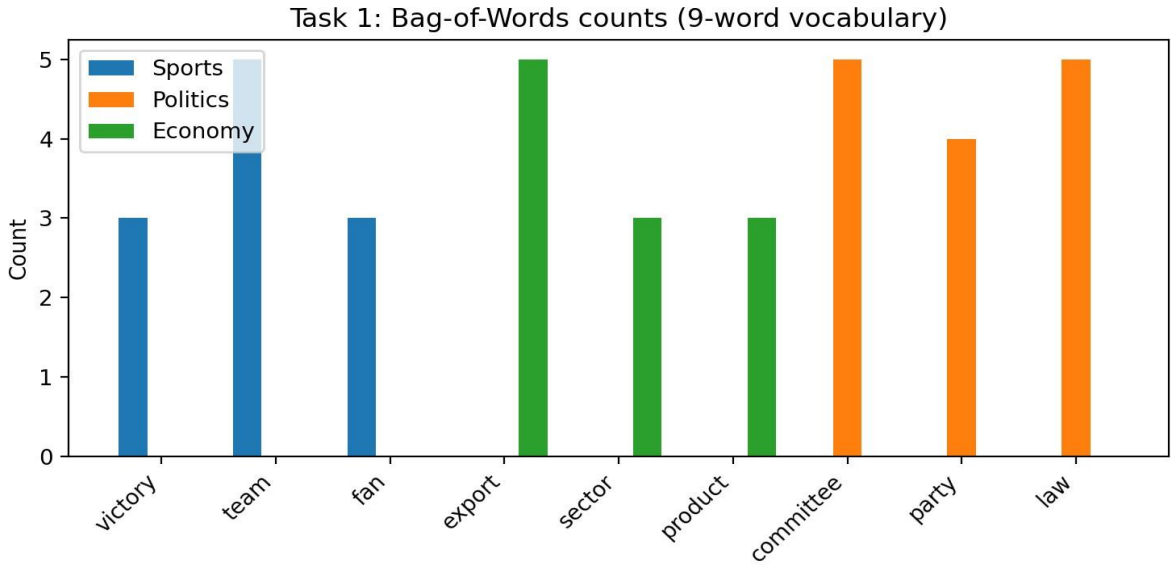
CENG 463 – Introduction to Machine Learning (HW2)

Task 1: Bag of Words (BoW)

Objective: Using the given 9-word vocabulary, compute 1×9 frequency vectors for three documents (Sports, Economy, Politics) and visualize the results.

BoW frequency vectors (rows: documents, columns: vocabulary terms):

Document	victory	team	fan	export	sector	product	committee	party	law
Sports	3	5	3	0	0	0	0	0	0
Politics	0	0	0	0	0	0	5	4	5
Economy	0	0	0	5	3	3	0	0	0



Discussion: The Sports document concentrates on sport-related tokens (victory/team/fan), Economy concentrates on export/sector/product, and Politics concentrates on committee/party/law. Because the vocabulary is fixed and small, any synonyms, inflections, or spelling variants that are not in the list are ignored. Therefore, the BoW representation is simple and interpretable, but it is sensitive to vocabulary choice and does not capture semantic similarity.

Task 2: Text Classification (Spam vs Normal)

Objective: Classify short messages as Spam or Normal using (i) Chi-square feature selection, (ii) TF-IDF representation, and (iii) KNN (k=3).

Step 1 – Chi-square feature selection

A chi-square test was applied to each candidate word to measure how strongly its presence correlates with the class label. The notebook shows a worked example for the word “free”. For this word, the contingency table is (Spam contains=3, Normal contains=0, Spam not=0, Normal not=4), which yields $\chi^2 = 7.0$. The top discriminative words selected were: free, win, meeting.

Step 2 – TF-IDF representation

Using the selected words, TF, IDF, and TF×IDF values were computed for each message to build the numerical feature matrix.

Word	IDF (log(N/df))
free	0.8473
win	1.2528
meeting	1.2528

TF-IDF feature matrix for training messages:

Message	Label	free	win	meeting
Msg1: Win free money now	Spam	0.2118	0.3132	0.0000
Msg2: Free prize winner announcement	Spam	0.2118	0.0000	0.0000
Msg3: Click here for free gift	Spam	0.1695	0.0000	0.0000
Msg4: Meeting scheduled for tomorrow	Normal	0.0000	0.0000	0.3132
Msg5: Project deadline next week	Normal	0.0000	0.0000	0.0000
Msg6: Lunch meeting at noon	Normal	0.0000	0.0000	0.3132
Msg7: Conference call tomorrow	Normal	0.0000	0.0000	0.0000

TF-IDF vectors for test messages:

Test Message	free	win	meeting
Test1: Free tickets available	0.2824	0.0000	0.0000
Test2: Team meeting tomorrow	0.0000	0.0000	0.4176
Test3: Winner congratulations prize	0.0000	0.0000	0.0000

Step 3 – KNN classification (k=3)

Each test TF-IDF vector was compared to all training vectors using Euclidean distance and Cosine distance; the predicted label is the majority class among the 3 nearest neighbors.

Predictions summary:

Test Message	Euclidean (k=3)	Cosine (k=3)
Free tickets available	Spam	Spam
Team meeting tomorrow	Normal	Normal
Winner congratulations prize	Normal	Spam

Discussion: “Free tickets available” is classified as Spam under both metrics because it is nearest to Spam messages containing the token “free”. “Team meeting tomorrow” is classified as Normal due to the strong “meeting” feature matching Normal training messages. “Winner congratulations prize” produces an all-zero TF-IDF vector because none of the selected features appear in the message. In this corner case, cosine similarity is undefined (zero vector norm), so the cosine-based prediction can differ from the Euclidean-based prediction. A practical mitigation is to expand the selected feature set (2–3 words may be too few) or to fall back to a default class when the feature vector is zero.

Task 3: Topic Modeling with BERTopic

Objective: Use BERTopic to categorize 20 Newsgroups documents from three categories (rec.sport.baseball, sci.space, sci.med), report topics, analyze top words, visualize, and test with new articles.

Step 1 – Dataset creation

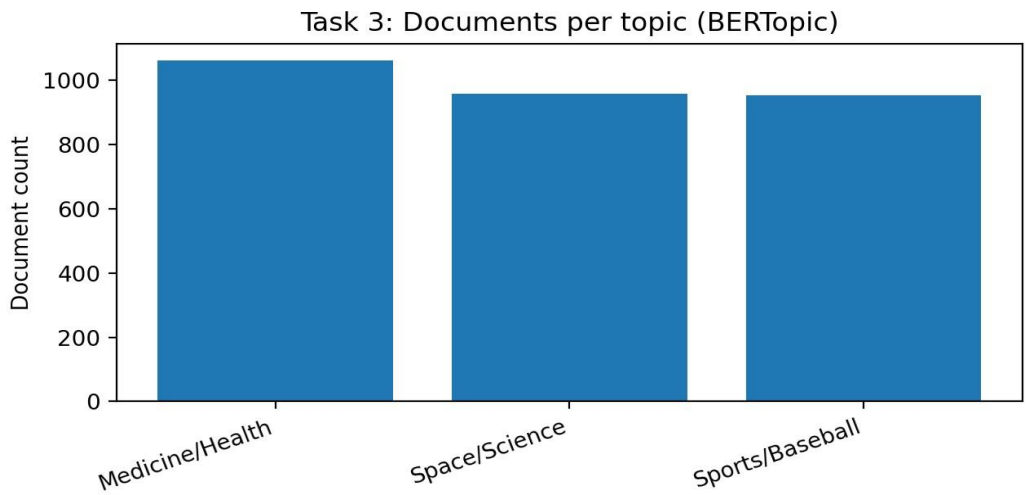
Dataset built using sklearn.datasets.fetch_20newsgroups with 3 selected categories. Total documents: 2,971 (rec.sport.baseball: 994; sci.med: 990; sci.space: 987).

Step 2 – Model training and topic summary

The notebook trains BERTopic using category labels to force exactly 3 topics (supervised mode). The model discovered 3 topics.

Topic summary and assigned labels:

Topic	IDCount	Model Name (auto)	Assigned Label	Top 5 words
0	1060	0_patients_treatment_hiv_aids	Medicine/Health	patients, treatment, hiv, aids, drug
1	958	1_launch_nasa_shuttle_mission	Space/Science	launch, nasa, shuttle, mission, satellite
2	953	2_player_pitching_stats_pitchers	Sports/Baseball	player, pitching, stats, pitchers,



Step 3 – Topic analysis

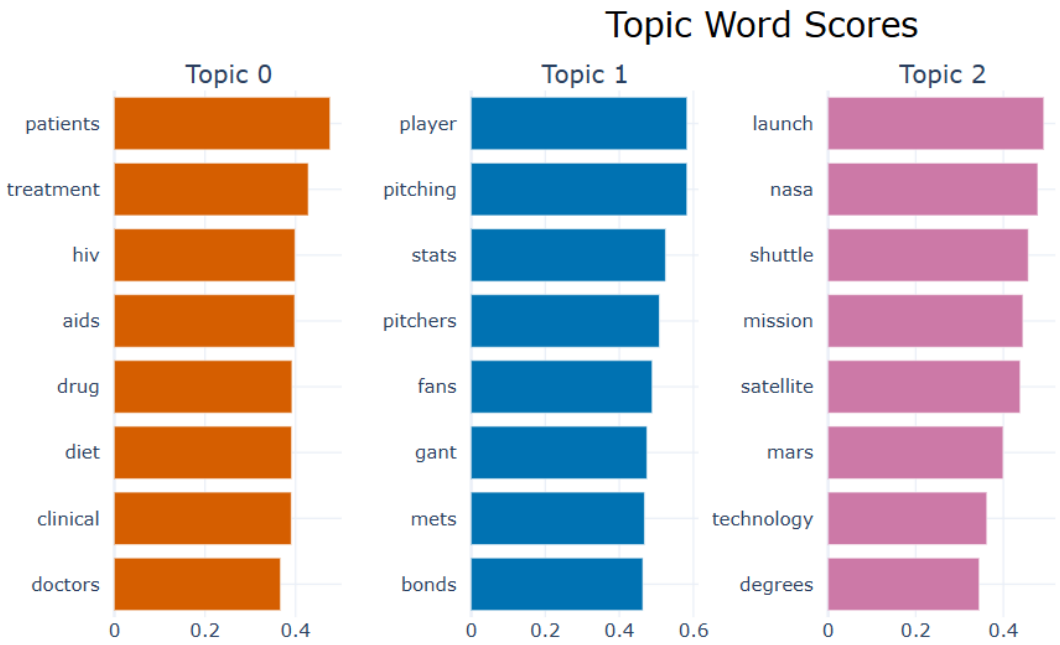
Top 5 words per topic are shown above. The topic-to-category alignment is strong: each topic corresponds to one of the three categories.

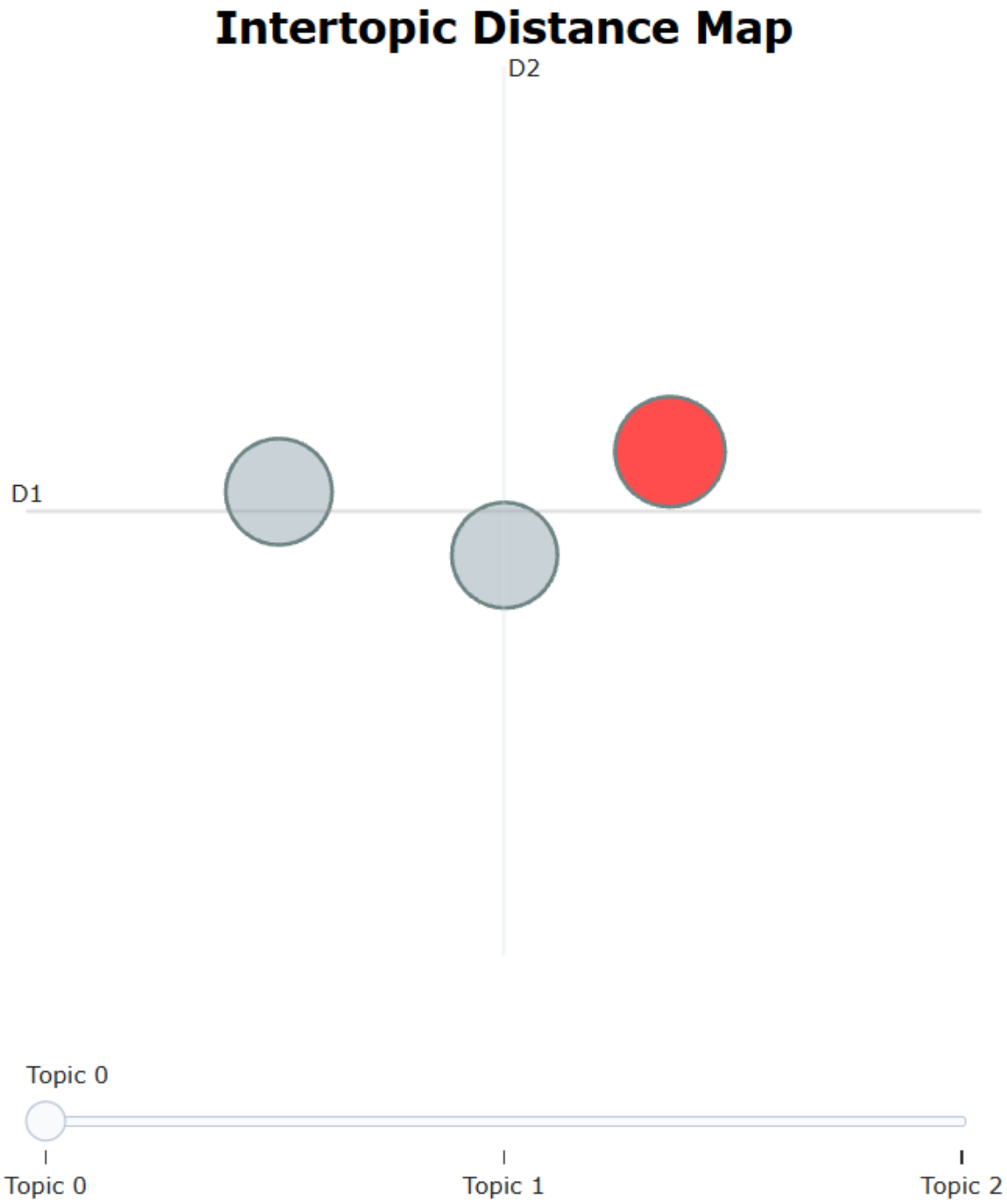
Document-topic distribution across true categories:

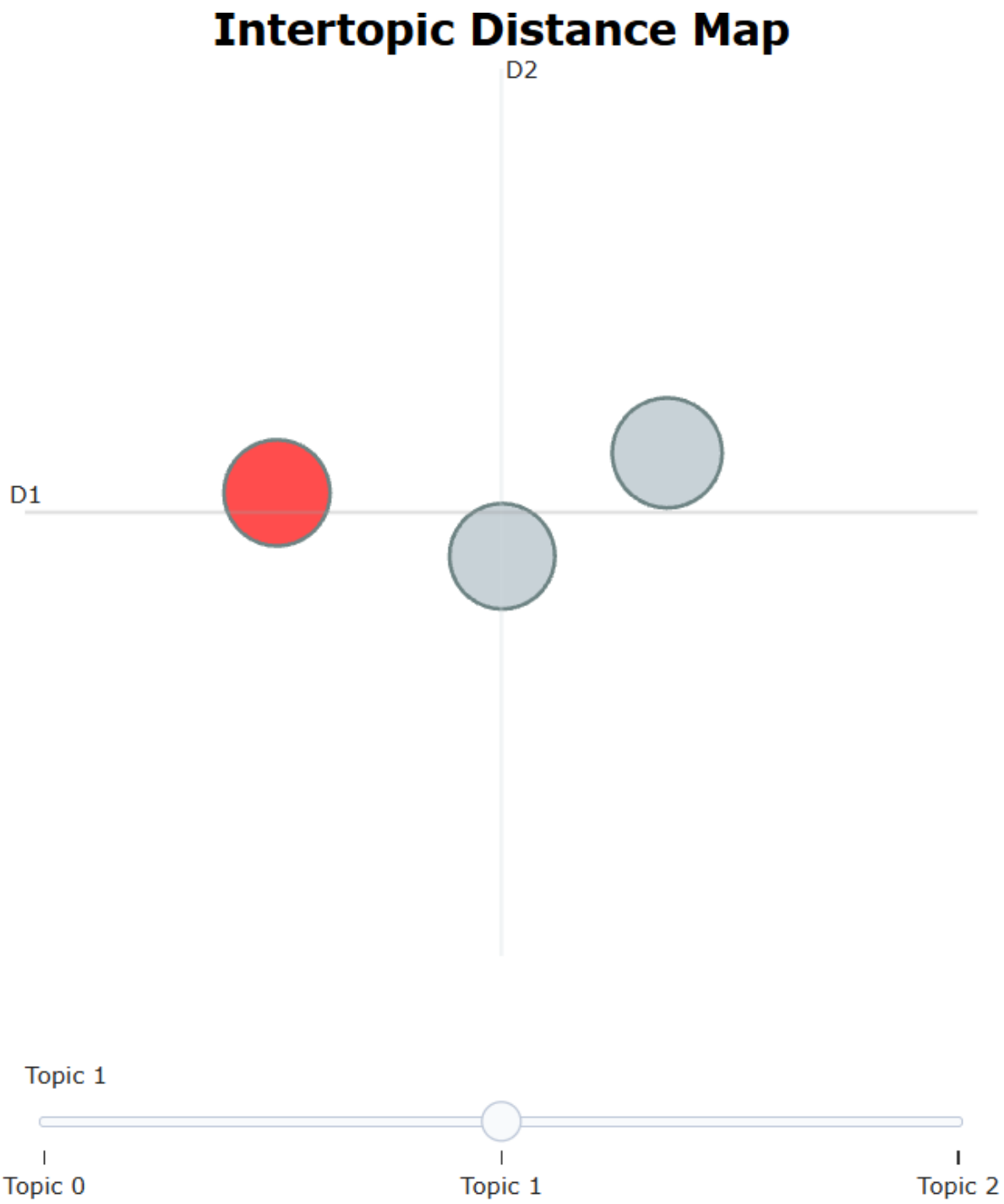
True Category	Medicine/Health	Space/Science	Sports/Baseball
rec.sport.baseball	40 (4.0%)	2 (0.2%)	952 (95.8%)
sci.med	986 (99.6%)	3 (0.3%)	1 (0.1%)
sci.space	34 (3.4%)	953 (96.6%)	0 (0.0%)

Step 4 – Visualization

In the notebook, BERTopic visualizations were generated: (i) a bar chart of top words per topic and (ii) a topic distance map showing topic similarities (interactive HTML produced by BERTopic).




Intertopic Distance Map
D2
D1
Topic 0
Topic 0
Topic 1
Topic 2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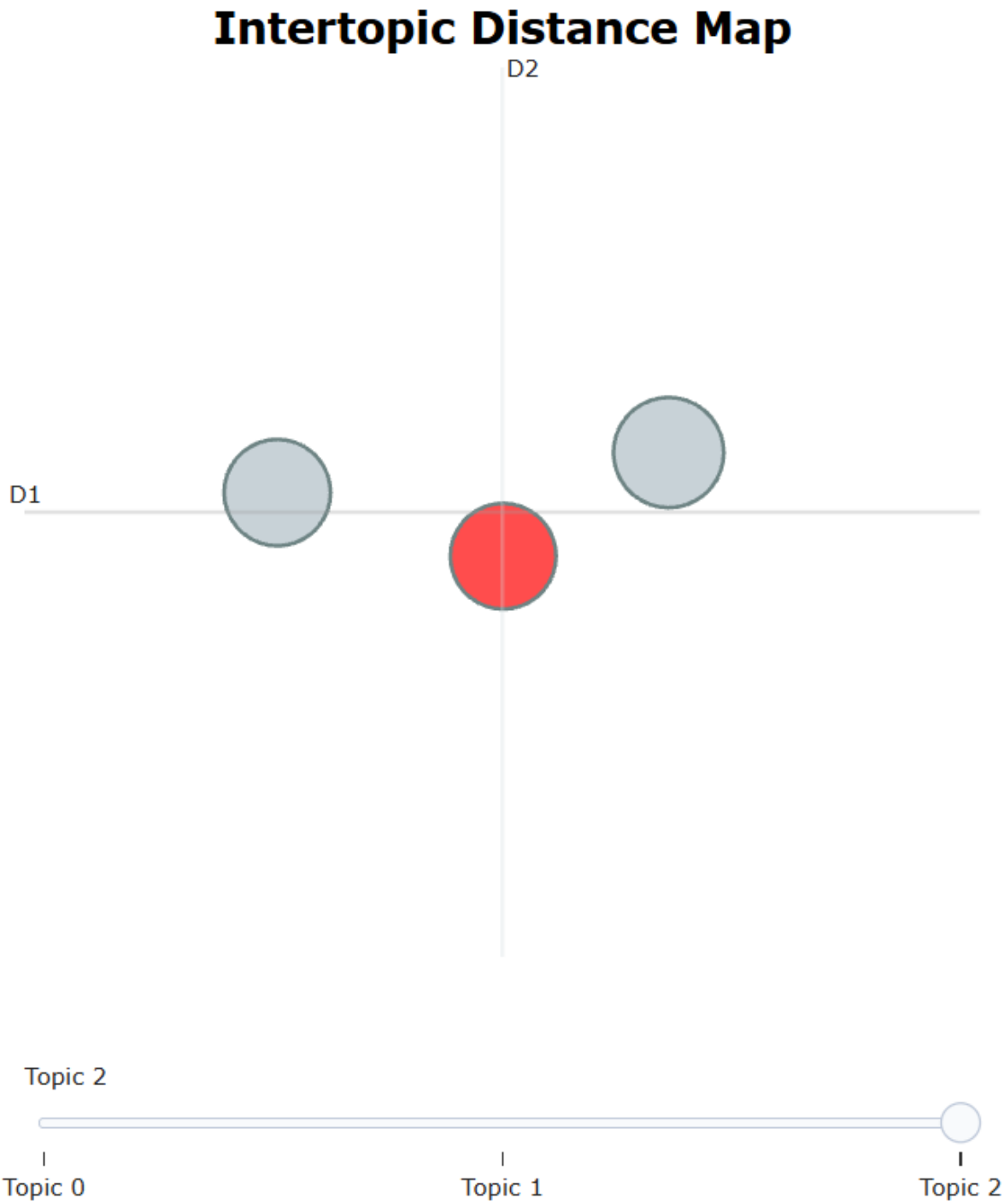

Step 5 – New document test

Three new articles (one per category) were written and topics were predicted using the trained model. All three predictions matched the expected category (3/3 = 100%).

Article	Expected	Predicted	Topic ID	Match
Article 1	Sports/Baseball	Sports/Baseball	2	Correct

Article 2	Space/Science	Space/Science	1	Correct
Article 3	Medicine	Medicine/Health	0	Correct

Discussion: Because the model was trained in supervised mode with known category labels, the resulting topics are highly aligned with the three target groups. This improves interpretability and prediction accuracy, but it reduces the model's ability to discover unexpected subtopics within each category. Topic quality and distances also depend on the embedding model and clustering hyperparameters.